

Security Lab GRC Documentation

MITRE ATT&CK Enterprise Matrix v18.1

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Created By: Paul Leone

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MITRE ATT&CK Enterprise Matrix v18.1

Document Control:

Version:	2.0
Last Updated:	April 2026
Owner:	Paul Leone
Framework Version:	ATT&CK v18.1 (October 2025)
Overall Coverage:	30% (65 techniques of 216)
Prior Version:	1.0: January 2026



Version 2.0: Change Log (from v1.0, January 2026)

Cloud content integrated inline into affected technique rows. Cloud node inventory: AWS (2 nodes), Azure (2 nodes), GCP (2 nodes): 6 nodes total via Tailscale WireGuard mesh VPN.

Tactic/Technique	Technique Name	Change	Status Impact
TA0001 T1199	Trusted Relationship	Recommendations updated: add cloud provider API gateway monitoring; AWS CloudTrail alerts on IAM role assumption anomalies; GCP SCC Asset API change detection; Azure Sentinel rule for cross-tenant federation events (Q2 2026).	PARTIAL
TA0005 T1562	Impair Defenses	Lab Implementation updated: cloud-native security service tampering detection added: CloudTrail alerts on GuardDuty disablement attempts; GCP audit logs capture SCC configuration changes; Azure Sentinel monitors Defender for Cloud policy modifications; all cloud security tool status monitored via Checkmk.	FULL
TA0010 T1537	Transfer Data to Cloud Account	Status upgraded from MINIMAL to PARTIAL. AWS S3 data access monitoring via CloudTrail (GetObject/PutObject); GCP Cloud Logging captures GCS bucket activity; Azure Monitor logs Storage account access; cloud-native DLP capabilities available but not formally deployed.	PARTIAL
TA0011 T1102	Web Service	Recommendations updated: GCP Cloud Armor application-level policy blocks suspicious web service patterns; AWS GuardDuty DNS anomaly detection covers DGA and tunneling patterns; Azure Sentinel KQL rules detect C2-indicative beaconing patterns in NSG flow logs.	MINIMAL

Overall Coverage Summary

Tactic	Coverage	Maturity	Priority Enhancements
Initial Access (TA0001)	67% (6/9)	Strong	SBOM tracking; content injection detection (T1659); cloud API gateway monitoring (T1199)
Execution (TA0002)	67% (9/13)	Strong	WMI logging; container CLI/API (T1059.013); container security enhancement
Persistence (TA0003)	37% (8/20)	Moderate	BITS monitoring; browser extension tracking; Office persistence; Python startup hooks (T1546.018)
Privilege Escalation (TA0004)	38% (5/13)	Moderate	Process injection detection; UAC bypass enhancement; container escape detection
Defense Evasion (TA0005)	21% (13/46)	Weak	Obfuscation analysis; pass-the-hash detection; alternate auth material monitoring; v18.1 techniques (T1678, T1679, T1036.012, T1562.013)
Credential Access (TA0006)	28% (4/15)	Weak	Kerberos monitoring; session hijacking detection; password store coverage
Discovery (TA0007)	32% (10/32)	Moderate	Share enumeration; peripheral tracking; local storage discovery (T1680) baseline
Lateral Movement (TA0008)	56% (5/9)	Strong	Pass-the-hash detection; WinRM logging; deployment tool monitoring
Collection (TA0009)	20% (4/18)	Weak	DLP implementation; email auditing; UEBA; database monitoring (T1213.006)
Exfiltration (TA0010)	46% (4/9)	Moderate	DLP deployment; web service monitoring; physical media auditing; cloud exfil (T1537 improved to PARTIAL)
Command & Control (TA0011)	42% (6/16)	Moderate	TLS inspection; remote access tool controls; web service C2 detection
Impact (TA0040)	46% (6/13)	Moderate	Disk modification monitoring; defacement detection; firmware integrity checks

Key Changes in ATT&CK v18.1

Change Type	Count	Details
New Techniques	12	T1059.013, T1213.006, T1678, T1546.018, T1562.013, T1680, T1036.012, T1677, T1681, T1679, T1518.002, T1204.005
Total Techniques	216	Increased from 191: 13% growth
Total Sub-techniques	475	Increased from 385: 23% growth
Lab Coverage Impact	-4%	Coverage adjusted from 34% to 30% due to denominator increase. T1537 improved from MINIMAL to PARTIAL with cloud monitoring additions.

Critical New Techniques Requiring Implementation

ID	Name	Gap / Action Required
T1059.013	Container CLI/API	Enable kubectrl exec and docker exec logging; Kubernetes audit log path configuration

T1678	Delay Execution	Detect extended sleep commands (>60s); correlate delays preceding malicious activity; sandbox evasion detection
T1679	Selective Exclusion	Monitor all security tool configuration changes; alert on antivirus exclusion path additions; implement security tool configuration baseline
T1036.012	Browser Fingerprint Masquerading	Deploy browser telemetry collection; implement user-agent anomaly detection; add browser fingerprinting signature detection
T1562.013	Network Device Firewall	Implement SNMP trap monitoring; add Checkmk SNMP monitoring for firewall config changes; deploy configuration backup validation
T1546.018	Python Startup Hooks	Implement Wazuh FIM on ~/.pythonrc, /usr/lib/python*/site-packages; monitor PYTHONSTARTUP environment variable
T1213.006	Data from Databases	Enable MySQL/PostgreSQL audit logs; deploy query monitoring for mass SELECT statements; add SIEM correlation for unusual database access patterns
T1680	Local Storage Discovery	Implement behavioral baseline for normal storage discovery; detect rapid enumeration (>10 queries/min); add cloud storage enumeration detection

Threat Intelligence Updates (v18.1)

ID	Name	Type / Action
G1053	Storm-0501	Threat Actor: Ransomware operations. Update MISP.
G1048	UNC3886	Threat Actor: VMware/network device exploitation. Update MISP.
G1052	Contagious Interview	Threat Actor: Supply chain via fake interviews. Update MISP.
G1051	Medusa Group	Threat Actor: Ransomware. Update MISP.
G1050	Water Galura	Threat Actor: APT/critical infrastructure. Update MISP.
S1240	RedLine Stealer	Malware: Credential theft. Update IDS signatures.
S1242	Qilin Ransomware	Malware: Ransomware. Update IDS signatures.
S1244	Medusa Ransomware	Malware: Ransomware. Update IDS signatures.
S1111	DarkGate Loader	Malware: Loader/dropper. Update IDS signatures.
S1229	Havoc C2 Framework	Malware: C2 framework. Update IDS signatures.
S1247	Embargo Ransomware	Malware: Ransomware. Update IDS signatures.
S1245	InvisibleFerret Backdoor	Malware: Backdoor. Update IDS signatures.

Technique Implementation Detail

TA0001 - Initial Access

ID	Technique Name	Status	Lab Implementation	Recommendations
T1078	Valid Accounts	FULL	Splunk correlation detects failed auth patterns; Wazuh monitors authentication logs; TheHive creates cases for	Deploy UEBA for behavioral baselines; implement

			brute force (>5 failures/5min); Authentik enforces account lockout; SSH tracks all authentication attempts	impossible travel detection
T1133	External Remote Services	FULL	pfSense logs all VPN connections; SSH session logging to SIEM; Traefik access logs track ForwardAuth requests; Uptime Kuma monitors service availability; Tailscale audit logs	Add geolocation-based alerts; implement time-of-day access policies
T1190	Exploit Public-Facing Application	PARTIAL	Weekly OpenVAS scans detect vulnerabilities; Nessus authenticated scans verify patch levels; Suricata/Snort IDS signatures; SafeLine WAF blocks OWASP Top 10 attacks (25% block rate)	Deploy virtual patching via WAF rules; implement honeypot services for early detection
T1566	Phishing	FULL	Shuffle phishing analysis workflow analyzes email headers, URLs, attachments; Cortex multi-engine analysis (VirusTotal, URLhaus, PhishTank); MISP correlates known phishing IOCs; Unbound blocks malicious domains	Add email gateway integration; deploy user reporting mechanism; implement DMARC/SPF/DKIM validation
T1091	Replication Through Removable Media	FULL	Group Policy blocks autorun.inf execution; Wazuh FIM monitors USB device insertion (Event ID 2003); alerts on unauthorized removable media; registry monitoring detects autorun changes	Deploy USB whitelisting; implement file encryption requirements for removable media
T1189	Drive-by Compromise	PARTIAL	Unbound blocks 1M+ malicious domains; Suricata/Snort detect exploit kit traffic patterns; SafeLine WAF protects web applications; DNS filtering prevents callback connections	Add browser isolation technology; implement content security policies; deploy endpoint isolation for high-risk browsing
T1199	Trusted Relationship	PARTIAL	MISP tracks vendor compromise campaigns; vulnerability scanning includes supply chain components; monitoring of third-party integrations	GAP: Implement vendor security assessments; deploy third-party risk management program; add API gateway

				monitoring; AWS CloudTrail alerts on IAM role assumption anomalies; GCP SCC Asset API change detection; Azure Sentinel rule for cross-tenant federation events (Q2 2026)
T1195	Supply Chain Compromise	PARTIAL	Package signature verification on Linux systems; Docker image verification (SHA-256); Git commit signing enforced	GAP: Deploy SBOM tracking (Trivy/Grype); implement dependency scanning; add software composition analysis
T1659	Content Injection	MINIMAL	Safeline WAF blocks common injection patterns; input validation at application layer	GAP: Deploy Content Security Policy (CSP) headers globally; implement subresource integrity (SRI) checks

Strengths: Strong authentication monitoring, phishing detection, network-based filtering

Gaps: Limited supply chain visibility, minimal content injection protection

Priority: Implement SBOM tracking (Q1 2026), enhance WAF rulesets (Q2 2026)

TA0002 - Execution

ID	Technique Name	Status	Lab Implementation	Recommendations
T1059.001	PowerShell	FULL	Sysmon captures PowerShell commands (Event ID 1, 4103, 4104); Wazuh detects obfuscated PowerShell (base64, encoded commands); Splunk searches for suspicious patterns (Invoke-Expression, DownloadString); script block logging enabled	Add PowerShell Constrained Language Mode; implement JEA (Just Enough Administration)
T1059.003	Windows Command Shell	FULL	Sysmon logs cmd.exe execution with full command-line; Wazuh detects living-off-the-land binaries (LOLBins); parent-child process analysis identifies anomalies; TheHive escalates suspicious executions	Deploy application whitelisting (AppLocker/WDAC); implement command-line auditing policies

T1059.006	Python	PARTIAL	Wazuh monitors Python process execution; auditd logs script execution on Linux; limited visibility into script content analysis	GAP: Implement Python script sandboxing; add runtime application self-protection (RASP); deploy code signing requirements
T1059.013	Container CLI/API	PARTIAL	Docker audit logging enabled; Kubernetes audit logs to SIEM; monitoring of kubectl/docker exec commands	GAP: Enable comprehensive container command auditing: kubectl exec logging, docker exec logging, container escape detection; enable Kubernetes audit logging (--audit-log-path); add Wazuh rules for container command execution
T11053	Scheduled Task/Job	FULL	Wazuh monitors scheduled task creation (Event ID 4698, 106); Sysmon tracks Task Scheduler API calls; FIM detects changes to cron/Task Scheduler directories; Splunk correlates privileged task creation	Add baseline of legitimate scheduled tasks; implement approval workflow for new tasks
T1106	Native API	PARTIAL	auditd logs system calls on Linux; Sysmon captures CreateRemoteThread, VirtualAllocEx; limited direct API call visibility	GAP: Deploy endpoint detection with API hooking; implement process injection detection; add kernel-level monitoring
T1129	Shared Modules	FULL	Sysmon ImageLoad events (Event ID 7) track DLL loading; Wazuh alerts on unsigned/untrusted DLLs; code signing validation enforced; detection of DLL search order hijacking	Implement DLL whitelisting; add certificate pinning for critical modules
T1203	Exploitation for Client Execution	PARTIAL	Suricata/Snort IDS signatures detect known exploit patterns; vulnerability scanning identifies unpatched client software;	GAP: Add exploit mitigation features (EMET/Windows Defender Exploit Guard); implement

			Cortex analyzes suspicious files with VirusTotal	application sandboxing; deploy zero-day protection
T1204	User Execution	FULL	Sysmon tracks process creation from user-writable directories; Wazuh monitors execution of files from Temp/Downloads; Cortex analyzes executables with multi-engine scanning; VirusTotal integration for malware detection	Add email attachment sandboxing; implement user training simulation; deploy macro blocking policies
T1569	System Services	FULL	Wazuh monitors service creation/modification (Event ID 7045, 7036); Sysmon tracks service installation; FIM alerts on service binary changes; unauthorized service detection	Implement service whitelisting; add approval workflow for service changes
T1047	Windows Management Instrumentation	MINIMAL	Limited WMI event monitoring; Sysmon tracks WMI activity (Event ID 19-21)	GAP: Enable comprehensive WMI logging; implement WMI subscription monitoring; add WMI persistence detection
T1609	Container Administration Command	PARTIAL	Docker audit logging enabled; Kubernetes audit logs to SIEM; monitoring of kubectl/docker exec commands	GAP: Implement admission controllers; add runtime container security (Falco); deploy container network policies
T1648	Serverless Execution	NOT APPLICABLE	No cloud serverless infrastructure in scope	N/A

Strengths: Comprehensive process monitoring, script execution detection, service tracking

Gaps: Limited WMI visibility, minimal API call monitoring, container CLI/API monitoring incomplete (T1059.013)

Priority: Enhance WMI logging (immediate), implement container command auditing (Q1 2026)

TA0003 - Persistence

ID	Technique Name	Status	Lab Implementation	Recommendations
T1098	Account Manipulation	FULL	Active Directory logs track group membership changes (Event ID 4728, 4732, 4756); Wazuh monitors user attribute modifications; Splunk	Implement privileged access management (PAM); add approval workflow

			correlation detects privilege escalation patterns; TheHive escalates unauthorized changes	for admin group changes
T1136	Create Account	FULL	Wazuh monitors account creation (Event ID 4720 Windows, useradd Linux); Splunk alerts on new accounts created outside business hours; unauthorized account detection via baseline comparison	Add account provisioning workflow; implement automated account lifecycle management
T1543	Create or Modify System Process	FULL	Sysmon Event ID 7045 tracks service creation; Wazuh monitors systemd unit changes on Linux; FIM alerts on service binary modifications; unauthorized service detection	Deploy service creation approval workflow; implement signed service binary requirement
T1547	Boot or Logon Autostart Execution	FULL	Sysmon Event ID 13 monitors registry Run keys; Wazuh FIM tracks startup folders, /etc/init.d, systemd; detection of persistence via: Run keys, startup scripts, scheduled tasks, WMI subscriptions	Add autostart location baseline; implement startup item approval process
T1053	Scheduled Task/Job	FULL	Comprehensive monitoring (see TA0002); persistence-specific detection includes: task creation by non-admin users, tasks with SYSTEM privileges, tasks pointing to suspicious paths	Implement task scheduler hardening; add scheduled task inventory baseline
T1505	Server Software Component	PARTIAL	Wazuh FIM monitors web directories (/var/www, IIS wwwroot); Yara signatures detect common web shells (China Chopper, WSO, b374k); IDS signatures for web shell activity; limited PHP/ASP behavioral analysis	GAP: Deploy web shell-specific detection (NeoPI); implement web application firewall with virtual patching; add file integrity baseline for web assets
T1078	Valid Accounts	FULL	Behavioral analysis detects: unusual login times, geographic anomalies, impossible travel, privileged account usage patterns; Splunk dashboards track authentication trends	Deploy full UEBA capabilities; implement privileged access analytics; add session recording for admin accounts

T1546.018	Event Triggered Execution: Python Startup Hooks	MINIMAL	No monitoring of Python startup files (.pythonrc, sitecustomize.py, usercustomize.py); limited tracking of PYTHONSTARTUP environment variable; Wazuh FIM does not monitor Python site-packages directories	GAP: Implement Wazuh FIM on ~/.pythonrc, /usr/lib/python*/site-packages, /etc/pythonrc.d; monitor PYTHONSTARTUP environment variable changes; add auditd rules for Python startup file access
T1556	Modify Authentication Process	PARTIAL	Wazuh FIM monitors: /etc/pam.d, LSASS process memory, credential providers, SSH authorized_keys; limited detection of in-memory credential provider injection	GAP: Implement memory scanning for credential provider tampering; add LSASS protection; deploy credential guard on Windows
T1197	BITS Jobs	MINIMAL	Limited BITS job monitoring	GAP: Enable BitsAdmin logging (Event ID 59, 60, 61); implement BITS job enumeration; add suspicious BITS transfer detection
T1037	Boot or Logon Initialization Scripts	PARTIAL	FIM monitors common script locations (bashrc, profile, logon scripts); limited detection of obfuscated scripts	GAP: Add script content analysis; implement script whitelisting; deploy behavioral analysis for script execution
T1176	Browser Extensions	MINIMAL	No browser extension monitoring	GAP: Deploy browser extension inventory; implement extension whitelisting; add browser telemetry collection
T1554	Compromise Client Software Binary	PARTIAL	FIM monitors application directories; code signing validation detects unsigned binaries; limited binary integrity verification	GAP: Implement application whitelisting; add runtime binary verification; deploy EDR with process hollowing detection

Strengths: Strong account monitoring, registry persistence detection, service tracking

Gaps: Limited BITS monitoring, minimal browser extension visibility, no Office persistence detection

Priority: Enable BITS logging (immediate), deploy application binary monitoring (Q1 2026)

TA0004 - Privilege Escalation

ID	Technique Name	Status	Lab Implementation	Recommendations
T1068	Exploitation for Privilege Escalation	PARTIAL	Weekly OpenVAS + monthly Nessus scans identify kernel exploits; Sysmon monitors unusual child processes from vulnerable applications; Wazuh detects known exploit patterns; limited zero-day detection	GAP: Deploy exploit mitigation features (EMET, Windows Defender Exploit Guard); implement kernel patch verification; add memory exploit detection
T1134	Access Token Manipulation	PARTIAL	Sysmon Event ID 4648 (explicit credential logon), 4672 (special privileges assigned); detection of SeDebugPrivilege abuse; limited token impersonation monitoring	GAP: Implement comprehensive token manipulation detection; add Windows Defender Credential Guard; deploy API hooking for token APIs
T1548	Abuse Elevation Control Mechanism	PARTIAL	Sysmon detects UAC bypass attempts (registry keys, file system redirects); Wazuh monitors eventvwr.exe, fodhelper.exe abuse; limited coverage of all bypass techniques	GAP: Enable UAC monitoring (Event ID 4103); implement UAC configuration auditing; add detection for UIPI bypasses
T1053	Scheduled Task/Job	FULL	Comprehensive monitoring of privileged task creation; detection of task creation with SYSTEM privileges; alerts on tasks created by non-admin users; tracking of task modification	Deploy task creation approval workflow; implement privileged task baseline
T1078	Valid Accounts	FULL	Tracks privileged account usage patterns; monitors admin account logons outside business hours; detects lateral movement via admin accounts; TheHive escalates anomalous privileged access	Add privileged access management (PAM) solution; implement just-in-time (JIT) admin access
T1055	Process Injection	MINIMAL	Sysmon captures CreateRemoteThread, QueueUserAPC; limited visibility into process	GAP: Deploy advanced EDR with process injection detection; implement memory

			hollowing, reflective DLL injection	scanning; add API hooking for injection techniques
T1543	Create or Modify System Process	FULL	Covered under TA0003 Persistence	See Persistence section
T1574	Hijack Execution Flow	MINIMAL	FIM monitors DLL directories; limited DLL search order hijacking detection; no PATH hijacking monitoring	GAP: Implement DLL preloading detection; add PATH manipulation monitoring; deploy signed binary enforcement
T1484	Domain Policy Modification	FULL	Active Directory audit logs track GPO changes (Event ID 5136, 5137, 5141); Wazuh alerts on unauthorized GPO modifications; SIEM correlation detects privilege escalation via GPO	Implement GPO change approval workflow; add GPO baseline comparison

Strengths: Strong privileged account monitoring, GPO change detection, scheduled task tracking

Gaps: Minimal process injection visibility, limited UAC bypass coverage, no container escape detection

Priority: Deploy advanced EDR for process injection (Q2 2026), enhance UAC monitoring (immediate)

TA0005 - Defense Evasion

ID	Technique Name	Status	Lab Implementation	Recommendations
T1027	Obfuscated Files or Information	PARTIAL	Cortex Yara signatures detect packing, base64 encoding; limited entropy analysis; detection of certutil decode, PowerShell encoding	GAP: Deploy advanced entropy analysis; implement sandbox detonation; add machine learning-based obfuscation detection
T1036	Masquerading	PARTIAL	Sysmon tracks process path anomalies; Wazuh validates code signing; detection of processes running from unusual locations; limited filename spoofing detection	GAP: Implement process name whitelist; add parent-child process validation; deploy binary reputation scoring
T1036.012	Masquerading: Browser Fingerprint	MINIMAL	No browser fingerprinting detection; limited user-agent analysis in Traefik/Suricata logs	GAP: Deploy browser telemetry collection; implement user-agent anomaly detection; add

				browser fingerprinting signature detection (Canvas, WebGL); monitor for automation frameworks (Puppeteer, Selenium)
T1070	Indicator Removal	FULL	Windows Event ID 1102 alerts on log clearing; Wazuh monitors file deletion in critical directories; immutable SIEM logs prevent tampering; FIM tracks log file modifications	Add log integrity verification; implement write-once-read-many (WORM) storage for critical logs
T1112	Modify Registry	FULL	Sysmon Event ID 13 captures all registry modifications; Wazuh FIM monitors critical registry keys; baseline deviation detection; alerts on security-related key changes	Implement registry change approval workflow; add registry rollback capability
T1140	Deobfuscate/Decode Files	PARTIAL	Process command-line monitoring detects certutil, base64 commands; limited file content analysis post-decoding	GAP: Deploy file detonation sandbox; implement recursive unpacking analysis; add memory-based deobfuscation detection
T1202	Indirect Command Execution	PARTIAL	Sysmon tracks LOLBin usage (mshta, regsvr32, rundll32, wmic); Wazuh alerts on unusual parent-child relationships; detection of script execution via non-standard interpreters	GAP: Implement comprehensive LOLBin baseline; add behavioral analysis for indirect execution; deploy application control policies
T1222	File and Directory Permissions Modification	FULL	Wazuh FIM alerts on permission changes; auditd logs chmod/chown on Linux; icaccls/takeown monitoring on Windows; detection of privilege escalation via ACL modification	Add permission change approval workflow; implement least privilege validation
T1484	Group Policy Modification	FULL	Covered under TA0004 Privilege Escalation; comprehensive GPO change auditing	See Privilege Escalation section

T1562	Impair Defenses	FULL	Wazuh monitors: Windows Defender tampering (registry, services), firewall rule changes, security service stops, log deletion attempts; alerts on antimalware service termination. Cloud: CloudTrail alerts on GuardDuty disablement attempts; GCP audit logs capture SCC configuration changes; Azure Sentinel monitors Defender for Cloud policy modifications; all cloud security tool status monitored via Checkmk	Implement tamper protection; add service recovery automation; deploy security tool health monitoring
T1562.013	Disable or Modify Network Device Firewall	PARTIAL	pfSense logs firewall rule changes via syslog; limited SNMP monitoring for network devices	GAP: Implement SNMP trap monitoring for all network devices; add Checkmk SNMP monitoring for firewall config changes; deploy configuration backup validation; alert on unauthorized firewall rule deletions
T1564	Hide Artifacts	MINIMAL	FIM monitors hidden file creation; limited alternate data stream (ADS) detection; no VBA stomping detection	GAP: Enable ADS scanning; implement VBA source code extraction; add hidden process detection
T1678	Delay Execution	MINIMAL	Limited sleep/timeout command monitoring; Sysmon captures timeout.exe, sleep commands; no behavioral analysis for time-delay patterns	GAP: Implement detection for extended sleep commands (>60 seconds); add correlation for delays preceding malicious activity; deploy sandbox evasion detection
T1679	Selective Exclusion	PARTIAL	Wazuh monitors Windows Defender exclusion additions (registry HKLM\SOFTWARE\Microsoft\	GAP: Monitor all security tool configuration changes; alert on

			Windows Defender\Exclusions); limited detection of firewall/EDR exclusions	antivirus exclusion paths (C:\Windows, C:\Users); track pfSense/firewall bypass rule additions; implement security tool configuration baseline
T1006	Direct Volume Access	MINIMAL	Limited raw disk access monitoring	GAP: Deploy volume shadow copy monitoring; add raw disk access alerting; implement VSS change detection

Strengths: Strong log protection, registry monitoring, security tool tampering detection

Gaps: Minimal obfuscation analysis, limited indirect execution detection, no alternate authentication material monitoring

Priority: Deploy advanced obfuscation detection (Q2 2026), implement pass-the-hash monitoring (Q1 2026)

TA0006 - Credential Access

ID	Technique Name	Status	Lab Implementation	Recommendations
T1003	OS Credential Dumping	FULL	Sysmon Event ID 10 detects LSASS process access; Wazuh alerts on Mimikatz signatures; detection of SAM/SYSTEM file access; monitoring of credential dumping tools (pwdump, gsecdump)	Deploy Windows Defender Credential Guard; implement LSASS protection (RunAsPPL); add memory scanning for credential extraction
T1110	Brute Force	FULL	Multi-source correlation: failed logins from firewall, SSH, Authentik, RDP; Shuffle workflow creates TheHive cases for brute force (>5 failures/5min); account lockout enforcement; IP-based rate limiting	Add distributed brute force detection; implement CAPTCHA after failed attempts; deploy geolocation-based blocking
T1555	Credentials from Password Stores	PARTIAL	Wazuh FIM monitors browser credential directories (Chrome, Firefox); limited coverage of password manager access (KeePass, 1Password); no macOS Keychain monitoring	GAP: Implement comprehensive password store monitoring; add credential access behavioral analysis; deploy browser

				credential protection
T1552	Unsecured Credentials	PARTIAL	Wazuh FIM monitors sensitive directories (/root/.ssh, .aws, .kube); Yara signatures detect credential patterns in files; limited coverage of memory credential scanning	GAP: Deploy secret scanning tools (TruffleHog, GitLeaks); implement memory credential detection; add configuration file credential scanning
T1528	Steal Application Access Token	MINIMAL	Limited OAuth token monitoring	GAP: Implement OAuth token theft detection; add API token usage monitoring; deploy token expiration enforcement
T1557	Adversary-in-the-Middle	MINIMAL	TLS validation prevents MITM; limited ARP spoofing detection	GAP: Deploy ARP spoofing detection (ArpWatch); implement certificate pinning; add network anomaly detection for MITM
T1558	Steal or Forge Kerberos Tickets	MINIMAL	No Kerberos ticket monitoring (Event ID 4769, 4768)	GAP: Enable Kerberos event logging; implement golden/silver ticket detection; add Kerberos encryption downgrade monitoring
T1539	Steal Web Session Cookie	MINIMAL	No browser cookie theft monitoring	GAP: Implement session hijacking detection; add browser telemetry; deploy session binding (IP, User-Agent validation)

Strengths: Strong brute force detection, LSASS access monitoring, credential dumping detection

Gaps: Minimal Kerberos monitoring, limited password store coverage, no session hijacking detection

Priority: Enable Kerberos logging (immediate), implement session monitoring (Q1 2026)

TA0007 - Discovery

ID	Technique Name	Status	Lab Implementation	Recommendations
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T1087	Account Discovery	FULL	Command-line monitoring detects: net user, net group, whoami, id, getent; Wazuh tracks LDAP queries (Event ID 4662); Active Directory enumeration alerts; detection of account enumeration via SMB	Add baseline of legitimate discovery activity; implement honeypot accounts for detection
T1482	Domain Trust Discovery	FULL	Active Directory logs LDAP queries (Event ID 4662); Wazuh monitors nltest, dsquery commands; detection of trust enumeration; SMB session monitoring	Implement AD enumeration baseline; add honeypot OUs for detection
T1083	File and Directory Discovery	FULL	Sysmon/auditd tracks dir, ls, find commands; detection of recursive directory traversal; monitoring of file enumeration in sensitive directories; alerts on rapid file system scanning	Add file access behavioral baseline; implement honeypot files for detection
T1046	Network Service Discovery	FULL	Suricata/Snort detect port scans (SYN scans, full connects, UDP scans); pfSense logs scan activity; Splunk correlates reconnaissance patterns; detection of network sweeps	Deploy deception technology (honeypots); implement port scan response automation
T1049	System Network Connections Discovery	FULL	Sysmon Event ID 3 logs network connections; auditd tracks netstat commands; monitoring of connection enumeration; detection of network profiling	Add network connection baseline; implement connection anomaly detection
T1018	Remote System Discovery	FULL	Suricata/Snort detect network sweeps; detection of ping sweeps, ARP scans; monitoring of network reconnaissance tools (nmap, masscan); pfSense logs scanning activity	Deploy network segmentation alerting; add internal reconnaissance detection
T1082	System Information Discovery	FULL	Command-line monitoring tracks: systeminfo, uname, hostname, Get-ComputerInfo; detection of system profiling commands; monitoring of WMI queries for system info	Implement discovery command baseline; add system information honeypot values

T1016	System Network Configuration Discovery	FULL	Monitors ipconfig, ifconfig, ip addr, route, netsh commands; detection of network adapter enumeration; tracking of routing table queries	Add network configuration change alerting; implement network topology obfuscation
T1518.002	Backup Software Discovery	FULL	Sysmon/auditd tracks backup software processes; command-line monitoring detects backup enumeration queries; registry monitoring (HKLM\SOFTWARE\Backup); process monitoring for backup service queries (Get-Service, sc query)	Implement backup software inventory baseline; add behavioral detection for rapid backup enumeration; deploy honeypot backup configurations
T1680	Local Storage Discovery	PARTIAL	Command-line monitoring detects df, mount, lsblk, fsutil commands; auditd logs storage queries on Linux; limited behavioral baseline for storage enumeration patterns	GAP: Implement behavioral baseline for normal storage discovery; detect rapid enumeration (>10 queries/min); deploy SIEM correlation for unusual storage profiling patterns; add cloud storage enumeration detection
T1007	System Service Discovery	PARTIAL	Monitors sc query, systemctl, service commands; limited behavioral analysis	GAP: Implement service enumeration baseline; add service discovery anomaly detection
T1010	Application Window Discovery	PARTIAL	Limited GUI enumeration detection; process monitoring tracks tasklist, Get-Process	GAP: Implement GUI interaction monitoring; add screenshot detection; deploy user activity analytics
T1217	Browser Bookmark Discovery	PARTIAL	Wazuh FIM monitors browser profile access; limited content analysis of bookmarks	GAP: Deploy browser telemetry collection; add bookmark enumeration detection; implement user behavior analytics

T1124	System Time Discovery	PARTIAL	Limited time discovery monitoring (net time, w32tm)	GAP: Add time synchronization monitoring; implement time-based behavioral analysis
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Strengths: Strong network reconnaissance detection, comprehensive command-line monitoring, AD enumeration tracking

Gaps: Limited share enumeration detection, minimal peripheral discovery monitoring, no password policy query detection

Priority: Implement share enumeration monitoring (Q1 2026), add peripheral device tracking (Q2 2026), enhance local storage discovery detection (T1680)

TA0008 - Lateral Movement

ID	Technique Name	Status	Lab Implementation	Recommendations
T1210	Exploitation of Remote Services	PARTIAL	Suricata/Snort signatures detect known exploits (EternalBlue, BlueKeep); vulnerability scanning identifies exploitable services; limited zero-day detection	GAP: Deploy exploit detection via behavioral analysis; implement exploit mitigation features; add honeypot services
T1021.001	Remote Desktop Protocol	FULL	Suricata monitors RDP traffic; Wazuh tracks RDP sessions (Event ID 4624 type 10, 4625); Splunk correlates unusual RDP patterns; detection of RDP from unusual sources; monitoring of RDP tunneling	Implement RDP gateway with MFA; add session recording; deploy geographic restrictions
T1021.002	SMB/Windows Admin Shares	FULL	Suricata monitors SMB traffic; Event logs track admin share access (Event ID 5140, 5145); Wazuh alerts on lateral movement via PsExec, WMIC; detection of administrative tool usage	Deploy SMB signing enforcement; implement network segmentation; add admin share access baseline
T1021.004	SSH	FULL	Comprehensive SSH logging to SIEM; Wazuh monitors all SSH sessions; detection of: key-based auth anomalies, unusual SSH sources, privilege escalation via sudo; Splunk tracks SSH lateral movement patterns	Implement SSH certificate authority; add session recording; deploy jump host architecture
T1080	Taint Shared Content	PARTIAL	Wazuh FIM monitors shared directories; detection of file modifications on network	GAP: Implement share access behavioral baseline; add file integrity

			shares; limited content tampering detection	verification; deploy honeypot files on shares
T1563	Remote Service Session Hijacking	PARTIAL	Limited session hijacking detection; monitoring of unusual session activity	GAP: Implement session validation (IP, User-Agent binding); add session termination on anomaly; deploy privileged session monitoring
T1021.006	Windows Remote Management	MINIMAL	Limited WinRM monitoring (Event ID 4688 for wsmprovhost.exe)	GAP: Enable comprehensive WinRM logging (Event ID 6, 91, 168); implement WinRM usage baseline; add remote PowerShell detection
T1072	Software Deployment Tools	MINIMAL	No SCCM/deployment tool monitoring	GAP: Implement deployment tool activity monitoring; add unauthorized deployment detection; deploy approval workflow
T1550	Use Alternate Authentication Material	MINIMAL	No pass-the-hash/pass-the-ticket detection	GAP: Enable detection of: pass-the-hash (Event ID 4624/4625 type 9), pass-the-ticket (Kerberos anomalies), token theft; deploy Credential Guard

Strengths: Strong RDP/SMB/SSH monitoring, comprehensive authentication logging, lateral movement detection

Gaps: Minimal WinRM visibility, no pass-the-hash detection, limited deployment tool monitoring

Priority: Implement pass-the-hash detection (immediate), enable WinRM logging (Q1 2026)

TA0009 - Collection

ID	Technique Name	Status	Lab Implementation	Recommendations
T1560	Archive Collected Data	PARTIAL	Process monitoring detects: 7zip, WinRAR, tar, zip commands; Wazuh tracks archive creation in suspicious locations; limited behavioral analysis of data staging	GAP: Implement data staging detection via file access patterns; add archive content analysis;

				deploy DLP for sensitive data archiving
T1123	Audio Capture	PARTIAL	Limited microphone access monitoring; process monitoring detects audio recording tools	GAP: Implement audio device access controls; add microphone usage monitoring; deploy privacy controls
T1005	Data from Local System	PARTIAL	Wazuh FIM detects mass file access; limited data exfiltration staging detection; monitoring of file access in sensitive directories	GAP: Deploy DLP for sensitive file access; implement data classification; add UEBA for abnormal file access
T1039	Data from Network Shared Drive	MINIMAL	No dedicated network share access monitoring beyond authentication logs	GAP: Implement share access behavioral baseline; deploy file access auditing on shares; add DLP for share-based exfiltration
T1025	Data from Removable Media	PARTIAL	Wazuh monitors USB device insertion (Event ID 2003); limited file access monitoring from removable media	GAP: Implement removable media file access tracking; add DLP for USB data transfers; deploy device control policies
T1074	Data Staged	MINIMAL	Limited staging location monitoring; file creation in Temp directories tracked	GAP: Implement behavioral detection for data staging; add anomaly detection for large file consolidation; deploy honeypot staging directories
T1213.006	Data from Information Repositories: Databases	MINIMAL	No database-specific monitoring implemented; limited visibility into database query patterns or mass data extraction	GAP: Implement database access logging (MySQL/PostgreSQL audit logs); deploy query monitoring for mass SELECT statements; add SIEM correlation for unusual database access patterns;

				implement database exfiltration detection (large result sets, off-hours queries)
T1114	Email Collection	MINIMAL	No email access monitoring (Outlook PST, webmail)	GAP: Deploy email access auditing; implement mailbox audit logging; add email exfiltration detection

Strengths: Basic archive detection, USB device monitoring, file access tracking

Gaps: Minimal DLP capabilities, no email collection detection, limited staging detection

Priority: Deploy DLP solution (Q2 2026), implement email auditing (Q1 2026), add UEBA for collection detection (Q2 2026)

TA0010 - Exfiltration

ID	Technique Name	Status	Lab Implementation	Recommendations
T1020	Automated Exfiltration	PARTIAL	pfSense monitors bandwidth usage; Prometheus tracks outbound traffic volume; Suricata detects large data transfers; threshold alerts for unusual upload rates	GAP: Implement behavioral baseline for normal data transfers; add time-series anomaly detection; deploy DLP for automated exfiltration prevention
T1030	Data Transfer Size Limits	PARTIAL	Network monitoring detects chunked transfers; Splunk analyzes transfer patterns; limited detection of data obfuscation via chunking	GAP: Implement transfer pattern analysis; add detection for slow exfiltration; deploy statistical analysis of outbound data sizes
T1048	Exfiltration Over Alternative Protocol	PARTIAL	Suricata monitors non-standard protocols; pfSense logs unusual port usage; DNS tunneling detection via query analysis (Unbound); ICMP tunneling detection	GAP: Enhance DNS tunneling detection (entropy analysis, subdomain length); add steganography detection; implement protocol anomaly detection
T1041	Exfiltration Over C2 Channel	FULL	Suricata/Snort detect C2 beacon patterns; pfSense logs all outbound connections; MISP correlates known C2 IOCs; Cortex enriches suspicious IPs with	Deploy automated C2 blocking; implement beacon detection (jitter analysis); add domain generation

			threat intelligence; TheHive tracks C2 communications	algorithm (DGA) detection
T1052	Exfiltration Over Physical Medium	PARTIAL	USB device insertion monitored; limited file transfer tracking to removable media	GAP: Implement USB file transfer auditing; add DLP for removable media; deploy device control policies with allow/block lists
T1567	Exfiltration Over Web Service	MINIMAL	TLS inspection limited; no web service-specific exfiltration detection (Dropbox, Google Drive, Pastebin)	GAP: Deploy web service usage monitoring; implement cloud DLP; add web traffic categorization; monitor OAuth token usage
T1537	Transfer Data to Cloud Account	PARTIAL	AWS S3 data access monitoring via CloudTrail (GetObject/PutObject); GCP Cloud Logging captures GCS bucket activity; Azure Monitor logs Storage account access; cloud-native DLP capabilities available but not formally deployed	GAP: Deploy CASB for cloud service monitoring; implement cloud DLP; add sanctioned vs. unsanctioned cloud service detection
T1029	Scheduled Transfer	MINIMAL	Limited detection of scheduled exfiltration tasks	GAP: Correlate scheduled tasks with network activity; add time-based exfiltration pattern detection; implement job scheduler monitoring
T1011	Exfiltration Over Other Network Medium	MINIMAL	No Bluetooth/WiFi exfiltration monitoring	GAP: Implement wireless network monitoring; add Bluetooth device detection; deploy air-gapped network controls

Strengths: Strong C2 detection, DNS tunneling monitoring, bandwidth alerting

Gaps: Limited web service exfiltration detection, no physical medium file tracking, minimal cloud exfiltration monitoring

Priority: Deploy DLP capabilities (Q2 2026), implement web service monitoring (Q1 2026), add physical media auditing (Q1 2026)

TA0011 - Command and Control

ID	Technique Name	Status	Lab Implementation	Recommendations
T1071	Application Layer Protocol	FULL	Suricata/Snort analyze HTTP/HTTPS traffic patterns; pfSense logs all application protocols; Traefik access logs provide HTTP visibility; detection of C2 over HTTP/HTTPS, DNS, SMTP	Deploy TLS inspection for encrypted C2; implement user-agent analysis; add HTTP header anomaly detection
T1132	Data Encoding	PARTIAL	Suricata detects base64, hex encoding in traffic; limited entropy analysis; Splunk searches for encoded data patterns	GAP: Implement comprehensive entropy analysis; add multi-layer decoding detection; deploy ML-based encoding detection
T1001	Data Obfuscation	PARTIAL	Suricata monitors traffic anomalies; limited protocol obfuscation detection; detection of steganography signatures	GAP: Deploy steganography detection tools; implement protocol conformance validation; add traffic normalization analysis
T1568	Dynamic Resolution	FULL	Unbound/Technitium DNS logs all DNS queries; pfSense DNS logs forwarded to SIEM; Suricata detects DGA patterns; detection of: fast flux, domain generation algorithms, DNS tunneling; MISP correlates suspicious domains	Enhance DGA detection with ML models; implement NXDomain tracking; add DNS reputation scoring
T1573	Encrypted Channel	PARTIAL	Suricata analyzes TLS metadata (certificate details, cipher suites, JA3 fingerprints); limited TLS decryption capabilities; detection of unusual TLS patterns	GAP: Deploy TLS inspection infrastructure; implement certificate anomaly detection; add JA3/JA3S fingerprint threat intelligence
T1571	Non-Standard Port	FULL	Suricata/pfSense detect protocol-port mismatches (HTTP on 8443, SSH on 443); Splunk correlates unusual port usage; detection of port obfuscation; monitoring of all non-standard service ports	Add port usage behavioral baseline; implement service fingerprinting; deploy protocol conformance validation

T1095	Non-Application Layer Protocol	PARTIAL	Suricata monitors ICMP tunneling; pfSense logs raw socket connections; limited detection of custom protocols	GAP: Implement comprehensive ICMP analysis; add GRE/IPIP tunnel detection; deploy protocol anomaly detection
T1572	Protocol Tunneling	PARTIAL	Suricata detects SSH tunneling, DNS tunneling, VPN-over-DNS; limited detection of protocol encapsulation techniques	GAP: Deploy comprehensive tunneling detection; implement encapsulation pattern analysis; add behavioral baseline for tunneling
T1090	Proxy	FULL	pfSense logs proxy connections; Suricata detects SOCKS, HTTP proxy usage; detection of: internal proxy usage, unusual proxy chains, for exit nodes	Add proxy authentication enforcement; implement proxy reputation scoring; deploy allowlist for legitimate proxies
T1219	Remote Access Software	MINIMAL	Limited remote access tool detection (TeamViewer, AnyDesk, LogMeIn signatures in Suricata)	GAP: Deploy comprehensive remote access tool inventory; implement application control for remote access; add remote access behavioral baseline
T1105	Ingress Tool Transfer	PARTIAL	Monitors file downloads via HTTP/HTTPS; PowerShell DownloadFile detection; limited executable download analysis	GAP: Implement comprehensive download monitoring; add executable reputation checking; deploy sandbox for downloaded files
T1102	Web Service	MINIMAL	Limited detection of C2 over legitimate web services (Twitter, GitHub, Pastebin)	GAP: GCP Cloud Armor application-level policy blocks suspicious web service patterns; AWS GuardDuty DNS anomaly detection covers DGA and tunneling

				patterns; Azure Sentinel KQL rules detect C2-indicative beaconing patterns in NSG flow logs; implement web service categorization and anomaly detection
T1104	Multi-Stage Channels	MINIMAL	Limited multi-stage C2 detection	GAP: Implement C2 session correlation across channels; add multi-protocol C2 detection; deploy behavioral analysis for staged communication

Strengths: Strong DNS monitoring, proxy detection, non-standard port identification

Gaps: Limited TLS inspection, minimal remote access tool detection, no traffic signaling monitoring

Priority: Deploy TLS inspection (Q2 2026), implement remote access tool controls (Q1 2026), add web service C2 detection (Q2 2026)

TA0040 - Impact

ID	Technique Name	Status	Lab Implementation	Recommendations
T1485	Data Destruction	PARTIAL	Wazuh FIM detects mass file deletion (>50 files/min); monitoring of del, rm, Remove-Item commands; limited detection of secure wipe tools	GAP: Implement file deletion behavioral baseline; add volume shadow copy deletion detection; deploy immutable backups
T1486	Data Encrypted for Impact	FULL	Wazuh detects mass file modification (>50 files/min); Shuffle ransomware workflow monitors: file extension changes (.encrypted, .locked), ransomware note creation (README.txt), encryption tool execution; automated containment via Active Response; sub-30-minute MTTR	Deploy ransomware-specific behavioral detection; implement canary files for early detection; add automated network isolation
T1490	Inhibit System Recovery	PARTIAL	Wazuh FIM monitors backup file access; Event ID 524 (VSS deletion) alerts; limited vssadmin, wbadmin command detection	GAP: Implement comprehensive backup tampering detection; add volume shadow

				copy protection; deploy immutable backup verification
T1498	Network Denial of Service	PARTIAL	Suricata detects SYN floods, UDP floods; pfSense monitors bandwidth spikes; Prometheus alerts on connection exhaustion; limited distributed DDoS detection	GAP: Deploy comprehensive DDoS mitigation; implement rate limiting at firewall; add uplink provider DDoS protection
T1496	Resource Hijacking	FULL	Prometheus monitors CPU/memory anomalies; Grafana alerts on sustained high resource usage; Wazuh detects cryptomining processes (xmrig, cpuminer); detection of GPU mining activity; Checkmk infrastructure monitoring	Add cryptomining network pattern detection; implement process reputation scoring; deploy GPU usage monitoring
T1489	Service Stop	FULL	Wazuh monitors service stop events (Event ID 7036); systemd service failures tracked; critical service protection alerts; detection of security service termination; automated service recovery	Implement service stop approval workflow; add critical service health checks; deploy service dependency mapping
T1491	Defacement	PARTIAL	Wazuh FIM monitors web directories; limited web content integrity verification	GAP: Implement website content baseline; add automated defacement detection; deploy web application integrity monitoring
T1561	Disk Wipe	MINIMAL	Limited detection of disk wipe tools (diskpart, dd, shred)	GAP: Implement disk modification monitoring; add MBR/boot sector protection; deploy bootloader integrity verification
T1499	Endpoint Denial of Service	PARTIAL	Resource exhaustion monitoring; process CPU/memory limits; limited fork bomb detection	GAP: Implement process spawn rate limiting; add resource quota enforcement; deploy fork bomb detection

Strengths: Excellent ransomware detection, strong resource hijacking monitoring, service stop protection

Gaps: Limited disk wipe detection, minimal defacement monitoring, no firmware integrity verification

Priority: Implement disk modification monitoring (Q1 2026), add defacement detection (Q1 2026), deploy firmware integrity checks (Q3 2026)

Version History

Version	Date	Changes
1.0	January 2026	Initial release. ATT&CK v18.1 coverage: 30% (65/216 techniques).
2.0	April 2026	Cloud IaaS integration (AWS, Azure, GCP: 6 nodes). T1537 upgraded MINIMAL to PARTIAL. T1562 cloud content added. T1199 and T1102 recommendations updated with cloud-native controls. Change log added.